Standard Operating Procedure: Specimen Collection, Handling and Rejection

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Department	Pre-analytical (Sample Collection)	Status	Approved
Prepared by	Quality Manager	Reviewed by	Consultant Pathologist
Approved by	Laboratory Director	Distribution	All collection and processing staff

1. Purpose

To make sure every blood and urine specimen is collected, labelled, transported and accepted in a way that gives accurate results and keeps patients and staff safe.

2. Scope

Applies to all phlebotomists, nurses, front-office staff and sample processing staff at the main laboratory, collection centres and home collection services.

3. Responsibilities

Role	Responsibility
Phlebotomist	Patient identification, collection, labelling at the point of collection, correct order of draw
Front office	Request form check, patient preparation instructions, billing
Sample processing staff	Specimen acceptance or rejection, centrifugation, rejection log
Section supervisor	Daily review of the rejection log, staff training, corrective action

4. Definitions

- **Two patient identifiers:** full name and date of birth, or full name and lab registration number. Bed number is never an identifier.
- **Haemolysis:** rupture of red cells; serum or plasma looks pink to red.
- **Order of draw:** the fixed sequence in which tubes are filled to prevent additive carry-over (CLSI GP41).

5. Materials

Vacuum tube holder and needles (21-23 G), tourniquet, 70% isopropyl alcohol swabs, gauze, adhesive dressing, gloves, tubes listed in the test catalog (BIODNA test master), sharps container, specimen transport box with ice packs.

6. Procedure

6.1 Before collection

1. Check the request form is complete: patient name, age, gender, tests, referring doctor, date and time.
2. Confirm patient preparation. Fasting tests such as Fasting Blood Sugar (8-10 hours) and Lipid Profile (10-12 hours preferred) must not be collected if the patient has eaten.
3. Identify the patient using two identifiers by asking the patient to state them. Never ask "Are you Mr X?".
4. For HIV testing, confirm written informed consent is signed before collection.

6.2 Venipuncture

1. Perform hand hygiene and wear gloves.
2. Apply the tourniquet 7-10 cm above the site for no longer than 1 minute. Prolonged tourniquet time falsely raises potassium and proteins.
3. Clean the site with 70% alcohol and let it air-dry for 30 seconds. Do not touch the site again.
4. Insert the needle bevel-up at a 15-30 degree angle and fill tubes in the order of draw (6.3).
5. Release the tourniquet as soon as blood flow starts, remove the needle and apply pressure with gauze.
6. Dispose of the needle immediately in the white sharps container. Never recap needles.

6.3 Order of draw and tube mixing

Order	Tube	Additive	Gentle inversions	Common tests
1	Blood culture bottle	Culture media	Not applicable	Blood culture
2	Light blue top / Black top	3.2% / 3.8% sodium citrate	3-4	PT/INR, APTT, D-dimer / ESR
3	Gold top (SST)	Clot activator and gel	5	LFT, KFT, lipids, thyroid, vitamins, serology
4	Green top	Lithium heparin	8-10	Stat chemistry
5	Lavender top	K2 EDTA	8-10	CBC, HbA1c, blood group, peripheral smear
6	Grey top	Sodium fluoride / potassium oxalate	8-10	Blood glucose

Citrate tubes must be filled to the fill line (at least 90%). An underfilled citrate tube gives falsely prolonged clotting times and must be rejected.

6.4 Labelling

1. Label every tube in front of the patient, immediately after collection, never before.
2. The label must show full name, age/gender, lab registration number or barcode, date and time of collection and collector's initials.
3. Never relabel or correct a label on a tube that has already been collected. Recollect instead.

6.5 Transport and processing

1. Transport specimens upright in a closed, leak-proof box. Keep chemistry and serology samples at 2-8 °C with ice packs; keep coagulation and blood culture samples at room temperature.
2. Allow SST tubes to clot for 30 minutes before centrifuging. Centrifuge as per tube manufacturer (typically 1300 g for 10 minutes).
3. Separate serum for electrolytes within 2 hours of collection.
4. Urine samples must reach the lab within 2 hours, or be refrigerated at 2-8 °C for up to 24 hours.

7. Specimen acceptance and rejection criteria

Sample processing staff check every specimen on arrival. A specimen meeting any rejection criterion below is not tested.

Rejection criterion	Why it matters	Action
Unlabelled or mislabelled tube; name or ID does not match request form	Risk of reporting a result on the wrong patient	Reject. Never relabel. Request recollection.
Haemolysed sample (pink or red serum/plasma)	Falsely raises potassium, LDH, AST and others	Reject for affected tests. Request recollection. Note haemolysis on report if recollection is impossible and the pathologist approves.
Clotted EDTA or citrate sample	Invalid cell counts and clotting times	Reject. Request recollection.
Underfilled citrate tube (below 90% fill)	Wrong blood-to-anticoagulant ratio	Reject. Request recollection.
Wrong tube type for the test	Additive interferes with the test	Reject. Request recollection in the correct tube.
Insufficient volume	Test cannot be run or repeated	Reject or run priority tests after informing the doctor.
Sample older than its stability limit (see test catalog)	Analyte has degraded	Reject. Request recollection.
Leaking or broken container	Biohazard and possible contamination	Reject. Follow spill procedure in BIODNA-SOP-SAF-001.
Non-fasting sample for a fasting test	Results not interpretable	Reject or report with a non-fasting comment if the doctor agrees.

7.1 Rejection procedure

1. Record the rejection in the specimen rejection log: date, time, patient ID, test, reason, staff name.
2. Inform the collection centre or referring doctor within 1 hour by phone, and record who was informed.
3. Arrange recollection on priority. Do not discard the rejected sample until the supervisor has reviewed it.
4. The supervisor reviews rejection trends weekly. A rejection rate above 2% for any collection centre triggers retraining.

8. Safety

Treat every specimen as potentially infectious. Wear gloves and a lab coat. In case of a needle-stick injury or blood spill, follow BIODNA-SOP-SAF-001 immediately.

9. Records

Specimen rejection log (BIODNA-F-PRE-004), critical value register (BIODNA-F-ANA-002), consent forms for HIV testing. Retain for 2 years.

10. References

- CLSI GP41: Collection of Diagnostic Venous Blood Specimens.
- ISO 15189:2022 Medical laboratories: Requirements for quality and competence.
- NABL 112: Specific criteria for accreditation of medical laboratories.
- BIODNA-QM-REF-001: Biological Reference Intervals and Critical Values.

11. Revision history

Version	Date	Change	Approved by
3.0	2026-04-01	Added order-of-draw table, underfilled citrate rule and 2% rejection-rate trigger	Laboratory Director
2.0	2025-04-01	Added home collection transport requirements	Laboratory Director
1.0	2024-04-01	First issue	Laboratory Director